

A document-gated neural topic mixture improves motif utility but not enough annotation coverage

MS2LDA research checkpoint

21 August 2026

Abstract

We tested one minimal architecture change intended to close the paper-relevant chemical-motif gap between the current neural MS2LDA model and Tomotopy. The experiment kept seed 42, 1,000 topics, six CPU threads, frozen training and validation data, existing token routing and losses, and the fixed epoch-40 checkpoint. It multiplied aggregated token topic mass by the square root of existing whole-spectrum topic evidence before normalization. On leakage-filtered validation data, useful high-confidence motifs—topics associated at probability at least 0.5 with compound-balanced structure overlap score (SOS) at least 0.6—increased from 20 to 83, while Tomotopy produced 138. The candidate therefore closed 53.4 percent of the gap and passed the predeclared SOS, likelihood, and stability checks. However, MAG annotation coverage reached only 42.0 percent, below the required 49.6 percent. The candidate was rejected and its test split was not accessed. The result supports document-level gating as a useful direction, but not this architecture as the new paper model.

1 Question and frozen comparison

MS2LDA treats spectra as documents, fragments and neutral losses as words, and recurring word distributions as Mass2Motifs [1]. The current neural checkpoint has better likelihood and much faster direct inference than the established Tomotopy model, but fewer chemically annotated and useful high-confidence motifs. This experiment asks whether one document-level gate can close that motif gap without a new encoder, loss, or parameter sweep.

The comparison fixes seed 42, 1,000 requested topics, six CPU threads, the training/validation split, vocabulary, token features, deterministic initialization, optimization schedule, and final epoch 40. The development runner exposes no candidate test matrices or test records. MAG uses an existing hash-verified index that excludes exactly the union of validation and test compound connectivity keys.

2 Document-gated mixture

The current router produces sparse top-2 token assignments and existing whole-spectrum document logits. After count-weighted token aggregation, the candidate calculates

$$g_d = \text{softmax}(\text{document logits}/T), \quad \theta_d = \text{normalize}(\text{token mass} \times g_d^{0.5}).$$

The same calculation is used in training and inference. Multiplication cannot create topic support absent from token routing, and an empty spectrum retains the existing uniform fallback. Setting the new mixture weight to zero exactly recovers the previous calculation. The gate adds no learned parameter, encoder, or auxiliary loss.

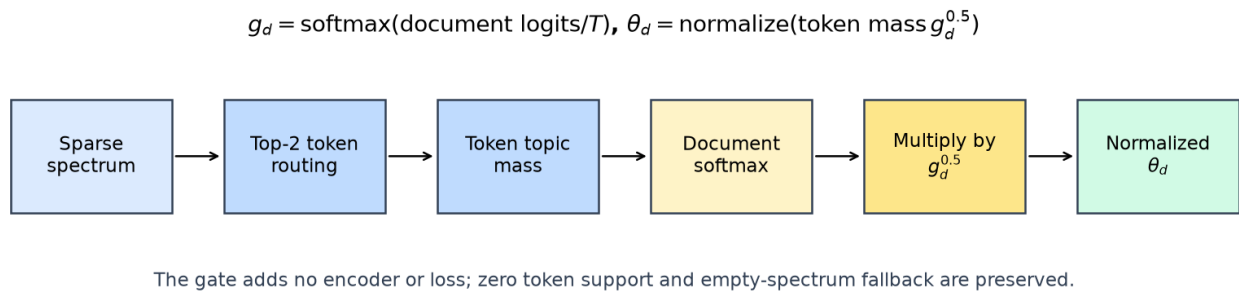


Figure 1: The minimal document-gated mixture. Existing document evidence reweights, but cannot expand, token-supported topic mass.

3 Paper-aligned validation

MAG annotation and SOS use validation spectra only. Library entries sharing a validation or test connectivity key are excluded before retrieval. Topic association follows the paper threshold, $\theta_{dk} \geq 0.5$. SOS is averaged over unique compounds within each topic. Eligible topics are divided into high (> 0.8), intermediate (0.6–0.8, inclusive), and low (< 0.6) bands.

The primary outcome is the number of high-confidence motifs with SOS at least 0.6. Acceptance required all of the following:

- closing at least 50 percent of the current-neural–Tomotopy useful-motif gap;
- MAG annotation coverage of at least 49.6 percent;
- mean high-confidence SOS no more than 0.02 below the current neural model;
- validation NLL no greater than 8.5266; and
- finite, stable training and evaluation.

The original experiment plan also proposed a 5,571-second fitting-time gate. After observing that the run was slow, the investigator explicitly demoted this condition to descriptive context. This post-run rule change is reported for transparency. It does not affect the scientific decision because the unchanged annotation-coverage gate rejects the candidate independently.

4 Results

Table 1: Validation-only paper-relevant comparison.

Metric	Current neural	Gated neural	Tomotopy
Optimized motifs	384	420	607
High-confidence SOS-evaluable motifs	42	156	206
SOS high (> 0.8)	10	33	58
SOS intermediate (0.6–0.8)	10	50	80
SOS low (< 0.6)	22	73	68
Useful high-confidence motifs (SOS ≥ 0.6)	20	83	138
Mean high-confidence SOS	0.6227	0.6373	0.6761
Median high-confidence SOS	0.5866	0.6185	0.6854
Model-fitting time (minutes)	84.4	177.8	76.0

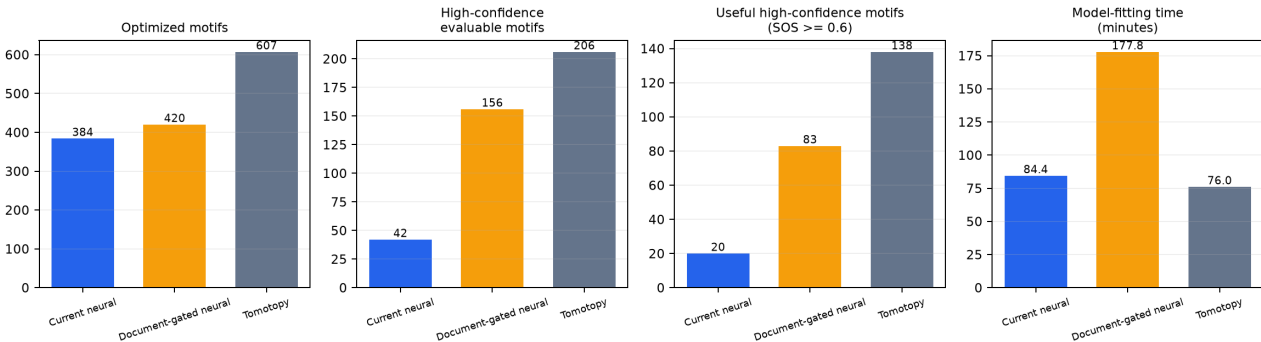


Figure 2: Headline paper-relevant outcomes and model-fitting time. The candidate improves all three motif counts but misses the annotation-coverage acceptance threshold.

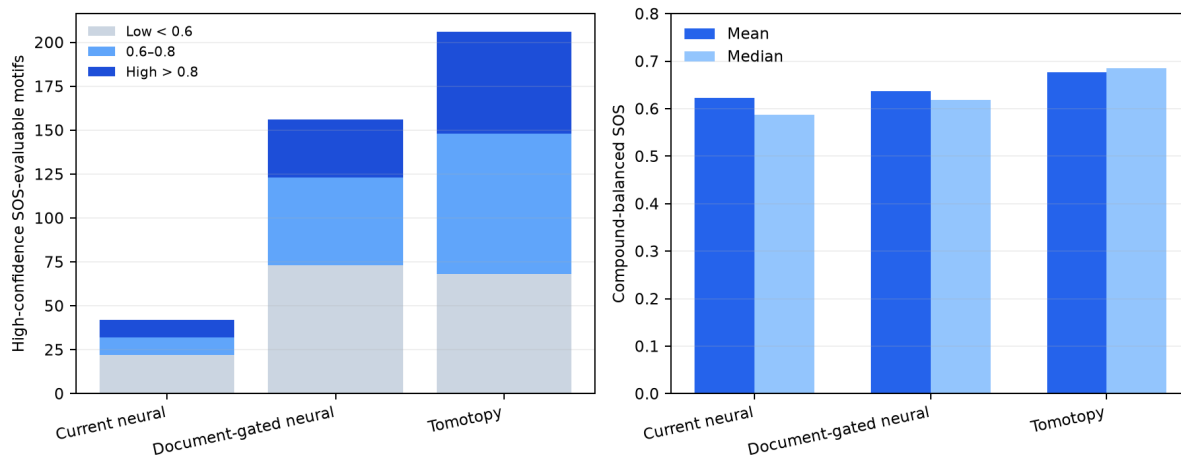


Figure 3: High-confidence chemical evidence. Left: SOS bands among evaluable motifs. Right: compound-balanced mean and median SOS.

The document gate increases high-confidence associated validation spectra from 87 to 582 and SOS-evaluable high-confidence motifs from 42 to 156. Tomotopy has 446 associated spectra and 206 evaluable motifs. Of those evaluable motifs, 20, 83, and 138 respectively have SOS at least 0.6. Thus the candidate recovers 63 of the 118 motifs in the original gap, or 53.4 percent, exceeding the 50 percent primary gate.

Quality among high-confidence motifs does not deteriorate. Candidate mean SOS is 0.6373 versus 0.6227 for the current neural model, and median SOS rises from 0.5866 to 0.6185. Tomotopy remains higher at mean 0.6761 and median 0.6854. Candidate validation NLL is 8.4166, 0.68 percent above the current 8.3594 but inside the 8.5266 safety bound. Training and validation remain finite and stable.

The failure is breadth of annotation. MAG optimizes 420 candidate motifs, compared with 384 for the current neural model and 607 for Tomotopy. Candidate coverage is therefore 42.0 percent: an improvement of 3.6 percentage points, but 7.6 points short of the predeclared 49.6 percent threshold. The architecture is rejected, the checkpoint hash is retained as evidence, and candidate test evaluation is not authorized.

Model fitting takes 177.8 minutes for the candidate, versus 84.4 minutes for the current neural run and 76.0 minutes for the reused Tomotopy source run. This is reported descriptively after the investigator removed runtime from acceptance; the candidate would have failed the original runtime rule as well.

5 Secondary inference context

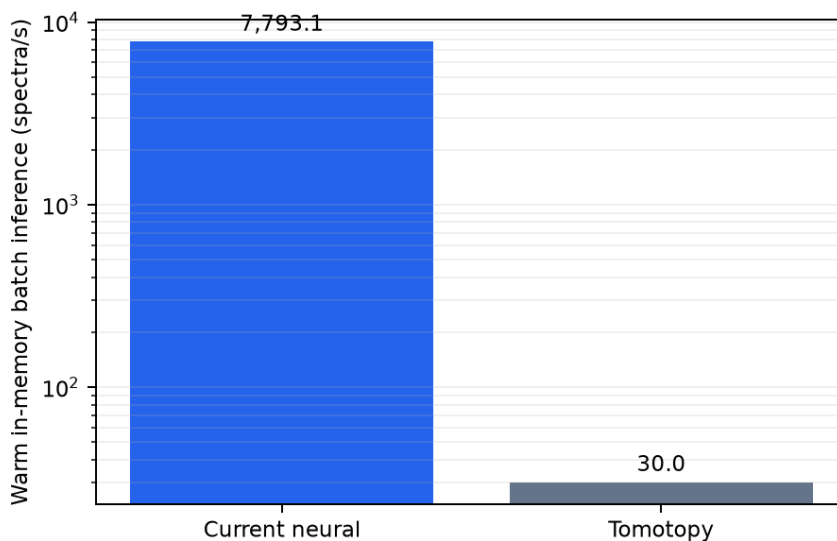


Figure 4: Previously measured warm in-memory batch inference for the accepted current neural model and Tomotopy. This secondary result is not a candidate acceptance gate; no candidate test latency was measured.

The earlier matched six-thread comparison measured 7,793 spectra/s for the current one-pass neural model

and 30.0 spectra/s for Tomotopy with 100 inference iterations, about 259-fold apart. “Warm in-memory batch inference” means that the model and batch are already loaded and one warm-up has completed. It is a real usable serving or repeated-batch scenario, but it excludes file loading, preprocessing, and first-call setup and is not end-to-end application latency.

6 Interpretation

The experiment confirms the frozen-checkpoint diagnosis: document evidence can convert diffuse mixtures into many more confident, chemically useful motifs. The improvement survives retraining and closes slightly more than half of the target gap. However, it does not make enough of the full 1,000-topic inventory annotatable, and it substantially increases fitting time. The likely next step is not a weight sweep on this rejected candidate. A new architecture should target broad motif annotatability directly while preserving the successful confidence mechanism and the current likelihood baseline.

This remains a single-seed research experiment. The fixed seed is deliberate: architecture development precedes multi-seed confirmation. No claim is made that the candidate replaces the production Tomotopy backend.

7 Reproducibility

The report retains the resolved comparison summary and hashed source manifests for candidate training, all three validation inference runs, all three MAG evaluations, and the final gate decision. The candidate checkpoint is fixed at epoch 40 with the following SHA-256:

```
8e52a88bce4d8138391f5898c2a3a16e  
65aaf3f4ec2d8b357851b1eeb70e1339
```

The run exposes no test matrix, and the report records that candidate test data were not accessed.

References

- [1] J. J. J. van der Hooft et al. “Topic modeling for untargeted substructure exploration in metabolomics.” *Proceedings of the National Academy of Sciences*, 2016. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5137707/>
- [2] M. Torres Ortega et al. “Large-scale discovery and annotation of substructure patterns in mass spectrometry profiles.” *Nature Communications*, 2026. <https://www.nature.com/articles/s41467-026-75038-0>